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Jitter reduction in mixed-signal processing circuits

Abstract

A signal processing circuit includes an analog front-end circuit and a digital delay circuit. The analog circuit receives a clock signal and provides a compensated clock signal. The digital delay circuit is coupled to the analog front-end circuit and provides a compensated sample clock signal in response to delaying the compensated clock signal. The analog circuit measures a variation of a power supply voltage and adjusts a gain through the analog circuit according to a measured variation of the power supply voltage.

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Background/Summary

FIELD

(1) This disclosure relates generally to signal processing circuits, and more specifically to jitter reduction in mixed-signal processing circuits.

BACKGROUND

(2) Noise in power supplies can affect signal transmission integrity in signal processing systems. For example, source synchronous clocking is a data transmission technique in which a source circuit generates and sends a clock signal along with a data signal, and a destination circuit uses the received clock signal to capture the data. Source synchronous clock forwarding input-output architectures include double data rate (DDR) dynamic random access memory (DRAM), low power DDR (LPDDR) DRAM, graphics DDR (GDDR) DRAM, and high bandwidth memory (HBM), as well as many other data transmission systems. For example, DDR5 memory uses data (DQ) and data strobe (DQS) signals for source-synchronous transmission during read and write transactions. On a write cycle, a host processor sends DQ signals and captures them on both edges of a data strobe signal (DQS). On a read cycle, the DDR5 memory sends DQ signals aligned on both edges of a data strobe signal (DQS). The host processor performs training to delay the launch of the data signal during writes so that the data arrives at the memory centered around each DQS transition, and performs training to internally delay the data signal during reads so that the data signal will be centered around each DQS transition. During reads and writes, the capturing device

has a period of time defined by the “data eye” in which it can reliably capture data. As DDR memory transmission speeds have increased, and can be as high as five Gigahertz (GHz) and higher, data eye closure has become a significant problem.

(3) One technique is to provide a receiver with continuous time linear equalization (CTLE). A typical receiver using this technique has a DQS path through an analog front end receiver with CTLE equalization, followed by programmable delay line, and then a distribution network that distributes the DQS signal to sample DQ signals on all data lanes, but this technique adds mismatch between the DQ and DQS paths.

(4) Since DQ and DQS paths are not matched, the mismatch creates relative jitter between DQ and DQS at the input of a sampler. Moreover, this jitter is dominated by un-correlated power supply noise caused by transmission of the DQ and DQS signals formed using the senders power supply, and reception of the DQS signal in an analog front-end circuit using another power supply with uncorrelated noise. The result is that unmatched power supply-induced jitter (PSIJ) can be as large as the width of the data eye itself, preventing reliable data reception. This problem is more difficult at current data rates that can be as large as 5 Gigabits per second (Gbps) and above.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates in block diagram form an exemplary data processing system that performs jitter reduction for signal processing circuits according to some embodiments;

(2) FIG. 2 illustrates in block diagram a signal processing circuit suitable for use in a source-synchronous clocking system according to some embodiments;

(3) FIG. 3 illustrates in schematic form an input stage suitable for use as the input stage of FIG. 1 according to some embodiments;

(4) FIG. 4 illustrates in partial block diagram and partial schematic form a gain stage suitable for use as the gain stage of FIG. 2 according to some embodiments;

(5) FIG. 5 illustrates in schematic form a feed-forward noise amplifier suitable for use as the feed-forward noise amplifier of FIG. 2 according to some embodiments;

(6) FIG. 6 illustrates in block diagram form a delay measurement circuit suitable for use with the feed-forward noise amplifier of FIG. 5 according to some embodiments; and

(7) FIG. 7 illustrates a timing diagram useful in understanding the operation of the delay measurement circuit of FIG. 6.

(8) In the following description, the use of the same reference numerals in different drawings indicates similar or identical items. Unless otherwise noted, the word “coupled” and its associated verb forms include both direct connection and indirect electrical connection by means known in the art, and unless otherwise noted any description of direct connection implies alternate embodiments using suitable forms of indirect electrical connection as well. The following Detailed Description is directed to electronic circuitry, and the description of a block shown in a drawing figure implies the implementation of the described function using suitable electronic circuitry, unless otherwise noted.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(9) A signal processing circuit includes an analog circuit and a digital delay circuit. The analog circuit receives a clock signal and provides a compensated clock signal. The digital delay circuit is coupled to the analog circuit and provides a compensated sample clock signal in response to delaying the compensated clock signal. The analog circuit measures a variation of a power supply voltage and adjusts a gain through the analog circuit according to a measured variation of the power supply voltage.

(10) A data processing system includes a data processor and a memory. The data processor has a memory controller and a physical interface circuit coupled to the memory controller. The memory

is coupled to the physical interface circuit over a memory bus. The memory bus conducts a data signal with a corresponding source-synchronous clock signal. At least one of the physical interface circuit and the memory includes a signal processing circuit. The signal processing circuit includes an analog circuit and a digital delay circuit. The analog circuit receives a clock signal and provides a compensated clock signal. The digital delay circuit is coupled to the analog circuit and provides a compensated sample clock signal in response to delaying the compensated clock signal. The analog circuit measures a variation of a power supply voltage and adjusts a gain through the analog circuit according to a measured variation of the power supply voltage.

(11) A method for compensation for power supply induced jitter of a clock signal in a signal processing circuit includes receiving the clock signal and providing a compensated clock signal in response to receiving the clock signal by an analog circuit. A compensated sample clock signal is provided in response to delaying the compensated clock signal. A data signal is sampled in response to an edge of the compensated sample clock signal. A variation of a power supply voltage used by the source-synchronous clock system is measured. A gain used in providing the compensated sample clock signal is adjusted in an amount proportional to a measured variation of the power supply voltage.

(12) FIG. 1 illustrates in block diagram form an exemplary data processing system **100** that performs jitter reduction for signal processing circuits according to some embodiments. Data processing system **100** includes generally a data processor **110** and a memory **120** connected to data processor **110** over a memory bus **140**.

(13) In some embodiments, data processor **110** is implemented as a microprocessor with one or more data processor cores. In some embodiments, data processor **110** is implemented as a system-on-chip (SoC) having multiple data processor cores, on-chip memory, and on-chip peripheral interfaces. Data processor **110** includes a memory controller **111** labelled “MC”, and a physical interface circuit **112** labelled “PHY” that is connected to memory controller **111** and conducts various signals over memory bus **140**. Shown in FIG. 1 are two signals of interest used in source-synchronous data transmission and reception, including a data bus conducting data signals labelled generically “DQ” over a set of conductors **141**, and true and complement data strobe signals labelled “DQS” and “DQS”, respectively, over a set of conductors **142**.

(14) Memory **120** includes a first set of integrated circuit terminals **121** for the DQ signals, and a second set of integrated circuit terminals **122** for the DQS and DQS signals. Memory **120** also includes a signal processing circuit **130** having an analog circuit **131** labelled “AFE”, a digital delay circuit **132**, and a sampling circuit **133**. Analog circuit **131** has an input connected to integrated circuit terminals **122** at nodes labelled “A”, and an output at nodes labelled “B”. Digital delay circuit **132** has an input connected to the output of analog circuit **131**, and an output at nodes labelled “C”. Samplers **133** have inputs connected to first set of integrated circuit terminals **121**, a capture input connected to the output of digital delay circuit **132**, and an output for providing a set of sampled data signals generically labelled “DQ.sub.IN”.

(15) In DDR memory systems, data is transmitted bidirectionally. During a read cycle, the memory transmits both the DQ signals and the DQS/DQS signals in a source-synchronous manner. During a write cycle, the data processor transmits the DQ signals and the DQS/DQS signals to the memory in a source-synchronous manner. Preferably, both data processor **110** and memory **120** use the jitter reduction technique and circuitry described herein when receiving data. However, either side can use it advantageously, and in data processing system **100**, the jitter reduction circuit is shown on the memory side only.

(16) Generally, analog circuit **131** operates as an analog front-end in a DQS receive path. It receives the DQS and DQS signals and compensates for a measured variation in power supply voltage by adjusting its gain using feed-forward gain modulation. The received clock signal, in this case data strobe signals DQS and DQS, goes through a digital delay, and compensation is performed for the whole path that is affected by power supply insertion jitter between nodes A and

C by overcompensating for the measured delay. The result is less jitter is better data transmission reliability from a larger data eye, and the capability of acceptable data transmission quality at higher data transmission speeds.

(17) FIG. 2 illustrates in block diagram a signal processing circuit **200** suitable for use in a source-synchronous clocking system according to some embodiments. Signal processing circuit **200** includes an analog circuit **210** and a digital delay circuit **220**. Analog circuit **210** includes an input stage **211**, a gain stage **212**, and a feed-forward noise amplifier **213**. Input stage **211** has inputs for receiving a differential data strobe signal including a first input for receiving the DQS signal and a second input for receiving the DQS signal, a control input for receiving a gain adjustment signal, and a differential output. Gain stage **212** has a differential input connected to the differential output of input stage **211**, a control input for receiving the gain adjustment signal, and a differential output. Feed-forward noise amplifier **213** has a power supply input for receiving the power supply voltage (not shown), and an output for providing the gain adjustment signal to the control input of input stage **211** and gain stage **212**. Digital delay circuit **220** has a differential input coupled to the differential output of gain stage **212**, and a differential output for providing the compensated clock signal as a differential signal including a true component DQS.sub.COMP, and a complementary component DQS.sub.COMP. Note that the data strobe signals clock data during a read or write cycle, and are considered clock signals even though they cease switching during inactive periods.

(18) Each of the circuits in signal processing circuit **200** has a power supply terminal for receiving a power supply voltage labelled “V.sub.DDQ” and a ground terminal for receiving a ground voltage labelled “V.sub.SSQ”, and forms a power supply voltage similarly referred to as V.sub.DDQ when V.sub.SSQ is at zero volts. Variations in the power supply voltage resulting from noise cause power supply induced jitter (PSU) on DQS and DQS. Feed-forward noise amplifier **213** compensates for this jitter by providing the gain adjustment signal. Moreover, in some embodiments, the gain adjustment signal overcompensates for the measured noise in order to compensate for additional PSU caused by digital delay circuit **230**, which also receives V.sub.DDQ. In some embodiments, the amount of overcompensation is calculated according to a measured or characterized ratio of analog delay to digital delay. In some embodiments, the overcompensation ratio can be determined by using the measured power supply voltage level to index into a lookup table with predetermined overcompensation ratios.

(19) The construction and operation of exemplary circuits that can be used for input stage **211**, gain stage **212**, and feed-forward noise amplifier **213** will now be explained with reference to exemplary circuits.

(20) FIG. 3 illustrates in schematic form an input circuit **300** suitable for use as input stage **211** of FIG. 2 according to some embodiments. Input circuit **300** includes a first current path **310**, a second current path **320**, a resistor **330**, and a capacitor **340**.

(21) First current path **310** includes transistors **311** and **312**, a capacitor **313**, and a resistor **314**. Transistor **311** is a P-channel metal-oxide-semiconductor (MOS) transistor having a source connected to V.sub.DDQ, a gate for receiving a bias signal labelled “Vbp”, a drain, and a feedback input. Transistor **312** is a P-channel MOS transistor having a source connected to the drain of transistor **311**, a gate for receiving a true input signal labelled “INP”, and a drain for providing a complement output signal labelled “OUTN”, and a feedback input. Capacitor **313** has a first terminal connected to the drain of transistor **312**, and a second terminal connected to V.sub.SSQ, and has an associated capacitance labelled “C.sub.P”. Resistor **314** is a variable resistor having a first terminal connected to the drain of transistor **312**, a second terminal connected to V.sub.SSQ, a feedback input, and has an associated resistance labelled “R.sub.D”.

(22) Second current path **320** includes transistors **321** and **322**, a capacitor **323**, and a resistor **324**. Transistor **321** is a P-channel MOS transistor having a source connected to V.sub.DDQ, a gate for receiving the Vbp signal, and a drain. Transistor **322** a P-channel MOS transistor having a source connected to the drain of transistor **321**, a gate for receiving a complement input signal labelled

“INN”, and a drain for providing a true output signal labelled “OUTP”. Capacitor **323** has a first terminal connected to the drain of transistor **322**, a second terminal connected to V.sub.SSQ, and has an associated capacitance C.sub.P. Resistor **324** is a variable resistor having a first terminal connected to the drain of transistor **322**, a second terminal connected to V.sub.SSQ, a feedback input, and has an associated resistance R.sub.D.

(23) Resistor **330** is a variable resistor having a first terminal connected to the source of transistor **311**, a second terminal connected to the source of transistor **321**, and a feedback input. Capacitor **340** has a first terminal connected to the source of transistor **311**, a second terminal connected to the source of transistor **321**, and a tuning input.

(24) Input circuit **300** uses CTLE to adjust the effective widths of transistors **311** and **321** and the values of resistors **314**, **324**, and **330**. Vbp is a bias signal for P-channel MOS transistors that sets the gain of input circuit **300**. When input circuit **300** is used as input stage **211**, Vbp receives the control signal from feed-forward noise amplifier **213** of FIG. 2, the INP and INN terminals receive the DQS and DQS signals, respectively, and the OUTP and OUTN signals are output signals provided to gain stage **212**.

(25) FIG. 4 illustrates in partial block diagram and partial schematic form a gain stage **400** suitable for use as gain stage **212** of FIG. 2 according to some embodiments. Gain stage **400** includes generally an input portion **410**, a true current mirror portion **420**, a buffer **430**, a complement current mirror portion **440**, and a buffer **450**.

(26) Input portion **410** includes transistors **411-415**. Transistor **411** is a P-channel MOS transistor having source connected to V.sub.DDQ, a gate for receiving the Vbp signal, a drain, and a feedback input used for CTLE. Transistor **412** is a P-channel MOS transistor having source connected to the drain of transistor **411**, a gate forming the positive input terminal receiving signal INP, and a drain. Transistor **413** is an N-channel MOS transistor having drain connected to the drain of transistor **412**, a gate, and a source connected to V.sub.SSQ. Transistor **414** is a P-channel MOS transistor having source connected to the drain of transistor **411**, a gate forming the negative input terminal receiving signal INN, and a drain connected to the gate of transistor **413**. Transistor **415** is an N-channel MOS transistor having drain connected to the drain of transistor **414**, a gate connected to the drain of transistor **412**, and a source connected to V.sub.SSQ.

(27) True current mirror portion **420** includes transistors **421-425**. Transistor **421** is an N-channel transistor having a drain connected to the drain of transistor **412**, a gate connected to the drain thereof, and a source connected to V.sub.SSQ. Transistor **422** is an N-channel transistor having a drain, a gate connected to the drain of transistor **421**, and a source connected to V.sub.SSQ. Transistor **423** is a P-channel transistor having a source connected to V.sub.DDQ, a gate, and a drain connected to the drain of transistor **422** and to the gate thereof. Transistor **424** is a P-channel transistor having a source connected to V.sub.DDQ, a gate, and a drain. Transistor **425** is an N-channel transistor having a drain connected to the drain of transistor **424**, a gate connected to the drain of transistor **412**, and a source connected to V.sub.SSQ.

(28) Buffer **430** has an input connected to the drains of transistors **424** and **425**, and an output forming the complement output signal OUTN.

(29) Complement current mirror portion **440** includes transistors **441-445**. Transistor **441** is an N-channel transistor having a drain connected to the drain of transistor **414**, a gate connected to the drain thereof, and a source connected to V.sub.SSQ. Transistor **442** is an N-channel transistor having a drain, a gate connected to the drain of transistor **441**, and a source connected to V.sub.SSQ. Transistor **443** is a P-channel transistor having a source connected to V.sub.DDQ, a gate, and a drain connected to the drain of transistor **422** and to the gate thereof and to the gate of transistor **424**. Transistor **444** is a P-channel transistor having a source connected to V.sub.DDQ, a gate connected to the drain of transistor **423**, and a drain. Transistor **445** is an N-channel transistor having a drain connected to the drain of transistor **444**, a gate connected to the drain of transistor **414**, and a source connected to V.sub.SSQ.

(30) Buffer **450** has an input connected to the drains of transistors **444** and **445**, and an output forming the true output signal OOTP.

(31) When used as gain stage **212** of FIG. 1, gain stage **400** receives the output of input stage **211** and provides a low impedance output capable of driving a significantly larger load. Transistor **411** operates as a current source when biased by signal Vbp. Transistors **412** and **414** operate as a differential pair that selectively divert the current from this current source based on the difference in voltage between the INP and INN terminals. When used with input stage **211**, INP and INN are connected to the OOTP and OUTN outputs of input stage **211**, and OOTP and OUTN are connected to digital delay circuit **220**.

(32) FIG. 5 illustrates in schematic form a feed-forward noise amplifier **500** suitable for use as feed-forward noise amplifier **213** of FIG. 2 according to some embodiments. Feed-forward noise amplifier **500** includes a capacitor **510**, a reference current generator **520**, a lowpass filter **530**, a transistor **540**, and a transistor **550**.

(33) Capacitor **510** has a first terminal connected to V.sub.DDQ, and a second terminal forming bias signal Vbp. Reference current generator **520** includes transistors **521-525**. Transistor **521** is a P-channel MOS transistor having a source connected to V.sub.DDQ, a gate connected to the second terminal of capacitor **510**, and a drain. Transistor **522** is an N-channel MOS transistor having a drain connected to the drain of transistor **521**, a gate connected to the drain thereof, and a source connected to V.sub.SSQ. Transistor **523** is an N-channel MOS transistor having a drain, a gate connected to the drain of transistor **522**, and a source connected to V.sub.SSQ. Transistor **524** is a P-channel MOS transistor having a source connected to V.sub.DDQ, a gate, and a drain connected to the gates of transistors **521** and **524**. Transistor **525** is an N-channel MOS transistor having a drain connected to the second terminal of capacitor **510**, a gate connected to the drain of transistor **522**, and a source connected to V.sub.SSQ. Lowpass filter **530** includes a resistor **531** and a capacitor **532**. Resistor **531** has a first terminal connected to the second terminal of capacitor **510**, and a second terminal. Capacitor **522** has a first terminal connected to the second terminal of resistor **531**, and a second terminal connected to V.sub.SSQ. Transistor **540** is a P-channel MOS transistor having a source connected to V.sub.DDQ, a gate connected to the first terminal of capacitor **532**, and a drain connected to the second terminal of capacitor **510**, and operates as a transconductance amplifier. Transistor **550** is a P-channel MOS transistor having a source connected to V.sub.DDQ, a gate connected to the second terminal of capacitor **510**, and a drain connected to the second terminal of capacitor **510**, and operates as a transconductance amplifier.

(34) In general, feed-forward noise amplifier **500** forms bias signals for P- and N-channel transistors, namely Vbp and Vbn, respectively. These bias signals include a gain adjustment component to compensate the gain for variations in power supply voltage V.sub.DDQ with respect to ground voltage V.sub.SSQ. Feed-forward noise amplifier **500** also provides overcompensation to account for the approximate delay through the digital delay circuit **230**.

(35) Feed-forward noise amplifier **500** operates as follows. Reference current generator **520** includes a P-channel current mirror formed by transistors **524** and **521** connected to an N-channel current mirror formed by transistors **522** and **523** on the N-channel side. Assuming the P- and N-channel transistors are sized the same, the circuit stabilizes when the current I_R through transistors **521** and **522** equals the current through transistors **524** and **521**. In this case, the voltage on the gate of transistor **525** also causes transistor **525** to conduct current $I_{sub.REF}$. The circuit again balances when $I_{sub.REF}$ is conducted by the parallel combination of transistor **540** and transistor **550**.

Transistors **540** and **550**, however, are sized proportionately such that for a gain value x , transistor **550** conducts $x \cdot I_{sub.REF}$, and transistor **540** conducts current $(1-x) \cdot I_{sub.REF}$. The gates of transistors **540** and **550** are driven by different signals. The gate of transistor **550** is driven by the second terminal of capacitor **510**, which varies with the high-frequency noise components of V.sub.DDQ. However, the gate of transistor **540** is driven by the output of lowpass filter **530**, which filters out the high frequency noise component. Thus, as V.sub.DDQ increases due to high

frequency noise, transistor **550** becomes less conductive, and Vbp decreases.

(36) In the exemplary embodiment, the width of transistors **521**, **522**, and **524** are the same, and the width of transistor **523** is made equal to four times the width of transistor **521**. Therefore, the width of transistor **550** is equal to 4*the width of transistor **524**, and the width of transistor **540** is equal to (1-x)*the width of transistor **524**. A.sub.v is the feed-forward noise gain and is equal to:

$$(37) A_v = 1 + \frac{W(550)}{W(540)} \quad [1]$$

Bias signal Vbp can thus be expressed as:

$$(38) V_{bp} = V_{bp_dc} + A_v * V_{supply_noise} \quad [2]$$

(39) Since Vsupply_noise is induced from the V.sub.DDQ supply, but Av*Vsupply_noise is applied to the gate of P-channel transistors, then if Av=1, the power supply noise would be canceled.

However, there is additional variable delay through digital delay circuit **230** induced by power supply noise, and Av is advantageously made greater than 1 to overcompensate for the injected variation in delay by these additional digital circuits.

(40) FIG. **6** illustrates in block diagram form a delay measurement circuit **600** suitable for use with feed-forward noise amplifier **213** of FIG. **5** according to some embodiments. Delay measurement circuit **600** is used to set the amplification and determine the x and (1-x) values as will now be described.

(41) Delay measurement circuit **600** includes an SR flip-flop **610**, an SR flip-flop **620**, a variable gain buffer **630**, a delay path **640**, a set of D latches **650** associated with SR flip-flop **610**, a set of D latches **660** associated with SR flip-flop **620**, and a coefficient determination circuit **670**.

(42) SR flip-flop **610** has a set input connected to the “A” node, a reset input connected to the “B” node, and an output for providing a signal labelled “O.sub.1”. SR flip-flop **620** has a set input connected to the “A” node, a reset input connected to the “C” node, and an output for providing a signal labelled “O.sub.2”. Variable gain buffer **630** has an input for receiving a clock signal labelled “CK”, an output, and a gain control input.

(43) Delay path **640** includes a set of buffers **641-647** that delay the output of variable gain buffer **630** to determine individual delays individual inverters **641-647** that generate clock signals that can be used to measure delays. Buffer **641** has an input connected to the output of variable gain buffer **630**, and an output for providing a signal labelled “CK1”. Buffer **642** has an input connected to the output of buffer **641**, and an output for providing a signal labelled “CK2”. This pattern repeats until buffer **647**, that has an input connected to the output of buffer **646**, and an output for providing clock signal CK7.

(44) D latches **650** include D latches **651-658**. D latch **651** has a reset input labelled “RST” for receiving a signal likewise labelled “RST”, a clock input connected to the output of variable gain buffer **630**, a D input for receiving the O1 signal, and an output for providing a signal labelled “d[0]”. D latch **652** has a reset input for the RST signal, a clock input connected to the output of buffer **641**, a D input for receiving the O1 signal, and an output for providing a signal labelled “d[1]”. This pattern repeats until D latch **658**, which has a reset input for the RST signal, a clock input connected to the output of buffer **647**, a D input for receiving the O1 signal, and an output for providing a signal labelled “d[7]”.

(45) D latches **660** include D latches **661-668**. D latch **661** has a reset input for receiving the RST signal, a clock input connected to the output of variable gain buffer **630**, a D input for receiving the O2 signal, and an output for providing a signal labelled “e[0]”. D latch **662** has a reset input for the RST signal, a clock input connected to the output of buffer **641**, a D input for receiving the O2 signal, and an output for providing a signal labelled “e[1]”. This pattern repeats until D latch **668**, which has a reset input for the RST signal, a clock input connected to the output of buffer **647**, a D input for receiving the O2 signal, and an output for providing a signal labelled “e[7]”.

(46) Coefficient determination circuit **670** has a first input for receiving the e[7:0] signals, a second input for receiving the input for receiving the e[7:0] signals, a first output for providing the value x,

and a second output for providing the value $(1-x)$.

(47) As noted above, the feed forward noise amplification factor

(48) $A_v (A_v = \frac{1}{x})$

is proportional to:

$$(49) \frac{(\text{Totaldelay})AC\text{delay}}{(\text{Analogdelay})AB\text{delay}} = \frac{1' \sin e[7:0]}{1' \sin d[7:0]} \quad [3]$$

By determining the ratio of the AC delay to the AB delay, coefficient determination circuit **670** can determine the values of x and $(1-x)$ that compensate for PSIJ in DQS and DQS. Furthermore, the overcompensation ratios can be determined by a suitable technique such as one of those described above.

(50) FIG. 7 illustrates a timing diagram **700** useful in understanding the operation of the delay measurement circuit **600** of FIG. 6. In timing diagram **700**, the horizontal axis represents time in picoseconds (ps), and the vertical axis represents the amplitude of various signals in volts. Shown in FIG. 7 are various waveforms of interest grouped into three groups.

(51) A first waveform group includes waveforms **710-714**. Waveform **710** shows a reset pulse labelled "RESET". Waveform **711** shows clock signal CK. Waveform **712** shows the CK signal at node A. Waveform **713** shows the CK signal at node B. Waveform **714** shows the CK signal at node C.

(52) A second waveform group includes waveforms **720** and **721**. Waveform **720** shows the O.sub.1 signal, which represents the pulse width from the activation of the CK signal at node A until the activation of the CK signal at node B. Waveform **721** shows the O.sub.2 signal, which represents the pulse width from the activation of the CK signal at node A until the activation of the CK signal at node C.

(53) A third waveform group includes waveforms **730-737**. Waveform **730** shows the CK0 signal at the output of variable gain buffer **630**. Waveform **731** shows the CK1 signal at the output of buffer **631**. Waveform **732** shows the CK1 signal at the output of buffer **632**, and so on, until Waveform **737** shows the CK7 signal at the output of buffer **637**.

(54) The operation of delay measurement circuit **600** of FIG. 6 can now be described with reference to timing diagram **700**. The pulse width of O.sub.1 is proportional to delay between node A and node B, i.e., delay of analog components. O.sub.1 is set by rising edge at the start point of delay (i.e., node A), and is reset by the rising edge at the endpoint of delay (i.e., node B). The pulse width of O.sub.2 is proportional to delay between node A and node C (i.e., the total delay of the unmatched path). O.sub.2 is set by rising edge at the start point of the delay (i.e., node A), and is reset by rising edge at the end point of the delay (i.e., at node C).

(55) Signals $d[7:0]$ and $e[7:0]$ are generated by over-sampling O.sub.1 and O.sub.2, respectively, by the clock pulse with gradual delay increases. The number of ones in $d[7:0]$ and $e[7:0]$ are proportional to the delay through analog circuit **131** and the total delay to sampling circuit **133**, respectively. Note that care should be taken during clock tree construction, to make sure clock arrival times across the SR latches that are used to generate $d[7:0]$ and $e[7:0]$ are matched tightly.

(56) The delay through the digital delay circuit **132**, therefore, is inherently measured and is equal to the time from the high-to-low edge of O.sub.1 to the high-to-low edge of O.sub.2. Coefficient determination circuit **670** then determines the values of x and $(1-x)$ to overcompensate for the full delay from the unmatched path from node A to node C.

(57) Thus, a signal processing circuit architecture that compensates for PSIJ has been described. The signal processing circuit uses feed-forward PSIJ compensation to avoid the long settling time of analog delay circuits. It compensates for the distortion caused by PSIJ by inducing a signal that "jitters" in the opposite direction to nullify the jitter through the uncompensated path. It nullifies jitter in the whole unmatched path by overcompensating the jitter transfer function of the analog circuit, e.g., the AFE, by amplifying the PSIJ noise to bias voltages used in the AFE. It determines the overcompensation ratio using a digital flip-flop bank. Determination of the required

overcompensation ratio can be performed by a simple look-up table created by monitoring the power supply.

(58) While particular embodiments have been described, various modifications to these embodiments will be apparent to those skilled in the art. For example, a receiver as shown and described herein can be used in a variety of signal processing circuits, such as source-synchronous data systems, including DDR, LPDDR, and GDDR DRAMs as well as other well-known systems. The technique was described in terms of exemplary circuits used in the analog front-end, but in other embodiments other suitable circuits can be used. Moreover, a variety of techniques can be used to determine a suitable overcompensation ratio.

(59) Accordingly, it is intended by the appended claims to cover all modifications of the disclosed embodiments that fall within the scope of the disclosed embodiments.

Claims

1. A signal processing circuit, comprising: an analog circuit for receiving a clock signal and providing a compensated clock signal; and a digital delay circuit coupled to the analog circuit for providing a compensated sample clock signal in response to delaying the compensated clock signal, wherein the analog circuit measures a variation of a power supply voltage and adjusts a gain through the analog circuit according to a measured variation of the power supply voltage.
2. The signal processing circuit of claim 1, further comprising: a sampling circuit for sampling a data signal in response to an edge of the compensated sample clock signal.
3. The signal processing circuit of claim 1, wherein: the analog circuit overcompensates an effect of the variation of the power supply voltage on jitter induced by the power supply voltage in the analog circuit.
4. The signal processing circuit of claim 3, wherein: the analog circuit overcompensates the effect of the variation of the power supply voltage on the jitter induced by the power supply voltage in the analog circuit by selecting an amount proportional to the measured variation of the power supply voltage in response to both a delay through the analog circuit and a delay through the digital delay circuit.
5. The signal processing circuit of claim 1, wherein the analog circuit comprises: an input stage having a power supply input for receiving the power supply voltage, a signal input for receiving the clock signal, a control input for receiving a gain adjustment signal, and an output; and a feed-forward noise amplifier having a power supply input for receiving the power supply voltage, and an output for providing the gain adjustment signal in response to the measured variation of the power supply voltage.
6. The signal processing circuit of claim 5, wherein the feed-forward noise amplifier comprises: a capacitor having a first terminal for receiving the power supply voltage, and a second terminal forming the compensated clock signal; a reference current generator for forming a reference current based on a voltage on the second terminal of the capacitor; a lowpass filter having an input coupled to the second terminal of the capacitor, and an output; a first transconductance amplifier having an input coupled to the second terminal of the of the lowpass filter, and an output coupled to the second terminal of the capacitor; and a second transconductance amplifier having an input coupled to the second terminal of the capacitor, and an output coupled to the second terminal of the capacitor.
7. The signal processing circuit of claim 6, further comprising: a delay measurement circuit, for measuring a first delay between the clock signal and the compensated sample clock signal, wherein the analog circuit sets respective gains of the first transconductance amplifier and the second transconductance amplifier in response to the first delay.
8. The signal processing circuit of claim 7, wherein: the delay measurement circuit determines a second delay between the clock signal and the compensated clock signal, and a third delay between

the clock signal and the compensated sample clock signal, and sets gains of the first transconductance amplifier and the second transconductance amplifier in response to a difference between the third delay and the second delay.

9. A data processing system comprising: a data processor having a memory controller and a physical interface circuit coupled to the memory controller; and a memory coupled to the physical interface circuit over a memory bus, the memory bus conducting a data signal with a corresponding source-synchronous clock signal, wherein at least one of the physical interface circuit and the memory comprises a signal processing circuit, the signal processing circuit comprising: an analog circuit for receiving a clock signal and providing a compensated clock signal; and a digital delay circuit coupled to the analog circuit for providing a compensated sample clock signal in response to delaying the compensated clock signal, wherein the analog circuit measures a variation of a power supply voltage and adjusts a gain through the analog circuit according to a measured variation of the power supply voltage.

10. The data processing system of claim 9, wherein: the analog circuit overcompensates an effect of the variation of the power supply voltage on jitter induced by the power supply voltage in the analog circuit.

11. The data processing system of claim 10, wherein: the analog circuit overcompensates the effect of the variation of the power supply voltage on the jitter induced by the power supply voltage in the analog circuit by selecting an amount proportional to the measured variation of the power supply voltage in response to both a delay through the analog circuit and a delay through the digital delay circuit.

12. The data processing system of claim 9, wherein the signal processing circuit further comprises: a sampling circuit for sampling the data signal in response to an edge of the compensated sample clock signal.

13. The data processing system of claim 9, wherein the analog circuit comprises: an input stage having a power supply input for receiving the power supply voltage, a signal input for receiving a clock signal, a control input for receiving a gain adjustment signal, and an output; and a feed-forward noise amplifier having a power supply input for receiving the power supply voltage, an input coupled to the output of the input stage, and an output for providing the compensated clock signal in response to the measured variation of the power supply voltage.

14. The data processing system of claim 13, wherein the feed-forward noise amplifier comprises: a capacitor having a first terminal for receiving the power supply voltage, and a second terminal forming the compensated clock signal; a reference current generator for forming a reference current based on a voltage on the second terminal of the capacitor; a lowpass filter having an input coupled to the second terminal of the capacitor, and an output; a first transconductance amplifier having an input coupled to the second terminal of the of the lowpass filter, and an output coupled to the second terminal of the capacitor; and a second transconductance amplifier having an input coupled to the second terminal of the capacitor, and an output coupled to the second terminal of the capacitor.

15. The data processing system of claim 14, further comprising: a delay measurement circuit, for measuring a first delay between the clock signal and the compensated sample clock signal, wherein the analog circuit sets respective gains of the first transconductance amplifier and the second transconductance amplifier in response to the first delay.

16. The data processing system of claim 15, wherein: the delay measurement circuit determines a second delay between the clock signal and the compensated clock signal, and a third delay between the clock signal and the compensated sample clock signal, and sets gains of the first transconductance amplifier and the second transconductance amplifier in response to a difference between the third delay and the second delay.

17. A method for compensation for power supply induced jitter of a clock signal of a signal processing circuit, comprising: receiving the clock signal; providing a compensated clock signal in

response to receiving the clock signal by an analog circuit; providing a compensated sample clock signal in response to delaying the compensated clock signal; sampling a data signal in response to an edge of the compensated sample clock signal; measuring a variation of a power supply voltage used by the signal processing circuit; and adjusting a gain used in providing the compensated sample clock signal in an amount proportional to a measured variation of the power supply voltage.

18. The method of claim 17, further comprising: overcompensating an effect of the variation of the power supply voltage on jitter induced by the power supply voltage in the analog circuit.

19. The method of claim 18, wherein the overcompensating comprises: selecting the amount proportional to the measured variation of the power supply voltage in response to both a delay through the analog circuit and a delay between the compensated clock signal and the compensated sample clock signal.

20. The method of claim 17, wherein providing the compensated sample clock signal comprises: providing the compensated sample clock signal in response to delaying the compensated clock signal using a feed-forward noise amplifier having a power supply input for receiving the power supply voltage, and an output for providing the compensated clock signal in response to the variation of the power supply voltage.
